

Machine Learning Technologies Image Project 2026

Due date: 11th May 2026 by 11pm. This is the absolute latest date, I cannot give any extensions for any reason. You should try to submit earlier.

Some students may be asked to give a Viva to defend their assignment work. This will be after the submission date. You must attend if requested.

Worth 30% of your overall grade.

1 Introduction

- ✓ In this assignment you will take a collection of images and build a model that identifies the category of each image. You will then write up your results as an academic paper.
- ✓ I do not know in advance what results you will achieve. It is possible that a good model cannot be built from scratch with this data using the tools we currently know. If you get poor results that is fine - it is about the process and your interpretation of what the results tell you.
- ✓ You are going to build multiple CNN models and compare them, ultimately selecting the best one and justifying that choice.
- ✓ You may want to use Google Colab (<https://colab.google/>) for training, as some models will take a while to run.

2 Dataset Choices

- ✓ A link to each dataset is provided on Moodle. You must use the version provided there.

2.1 Garbage Classification

This dataset contains images of household waste across **10 categories**: metal, glass, biological, paper, battery, trash, cardboard, shoes, clothes, and plastic. All images are 384×384 pixels and in colour.

The dataset is derived from the Garbage Dataset (GD) [2], a publicly available benchmark for automated waste segregation. The images were collected through multiple methods including a mobile application and curated web sources, and exhibit realistic variability in lighting, background, and object occlusion – which makes the classification task genuinely challenging.

Approximate size: 400MB.

~~2.2~~ Food Classification

This dataset contains images of food across a number of categories, at 512×512 pixels. The dataset is a curated subset of the Food-101 dataset [1]. You will need to combine the files from food1.zip and food2.zip.

Approximate size: 1GB.

- ✓ **If you choose Food:** because training will take longer, you are still expected to explore the same range of models described below. Manage your time accordingly.

If you choose Garbage: because the dataset is smaller and faster to train, you are expected to explore a wider range of hyperparameter combinations and document the effect of each change more thoroughly.

3 What to Do

3.1 First Steps

✓ You will need a training set, a validation set **and** a test set. The test set is held back for final evaluation of your chosen model only – it is not used to select hyperparameters or to compare models.

✓ Use `image_dataset_from_directory` to load the data. For a dataset of this size this is the correct approach – if you are not using it, question why not.

✓ A suggested split is 70% training, 10% validation, 20% test, but you may choose differently. Whatever you choose, document it clearly in your paper and make sure your validation set is used only for model selection and your test set only for final reporting.

You must do this. Any random seeds must use your student ID number (without the G and leading zeros). For example, if your ID is G00432121 then your seed must be 432121. If you do not do this you will lose a significant number of marks.

3.2 Models

You should explore combinations of the following:

✓ • **Image size.** Smaller images train faster but may lose detail. Compare at least two sizes, or justify your choice of a single size. Ensure your chosen size is compatible with VGG16 for the transfer learning part.

✓ • **RGB colour vs greyscale.** Colour generally carries useful information, but does it here? Compare the two.

✓ • **Network depth.** Does adding more convolutional layers help?

✓ • **Overfitting.** What measures do you take? (Data augmentation, dropout, early stopping, regularisation – you should be considering these.)

You must also complete both of the following:

✓ 1. **CNN from scratch.** Define the architecture yourself, train it and interpret the results. Start with one model, diagnose whether overfitting or underfitting is occurring, adjust hyperparameters accordingly, and iterate until you are satisfied. Then build a greyscale version using the same hyperparameters for direct comparison.

✓ 2. **Transfer learning.** Use one of VGG16, ResNet50, or MobileNetV2 as your base model. There is no point trying greyscale with transfer learning - these models were trained on RGB images and expect three channels. After your initial transfer learning model is trained, perform **at least one epoch of fine-tuning**: unfreeze the top layers of the base model, use a very low learning rate (e.g. 10^{-5}), and compare the validation loss before and after. Report whether it helped.

A reasonable workflow is:

✓ • Get a working RGB CNN from scratch. Tune it until you are happy with it.

✓ • Build a greyscale version with the same hyperparameters and compare.

✓ • Build an RGB transfer learning model. Try a small number of hyperparameter changes if needed.

✓ • Fine-tune for one epoch and see if validation loss improves.

✓ • Use the validation results to pick your overall best model, then evaluate once on the test set.

3.3 Metrics

Use the standard metrics to evaluate and compare models (accuracy, precision, recall, F1-score). In addition:

- Include a **confusion matrix** for your final chosen model. Which categories does it confuse most? Is there a pattern?
- Include **training time** and **inference time** in your comparisons. Use `timeit` for inference measurements on the test set. Do not use `timeit` on training.
- Include **parameter count** (from `model.summary()`) for each model. Compare the size of your scratch model to the transfer learning model. Consider what this means in practice.

3.4 Class Imbalance

Inspect the class distribution as part of your initial data analysis. If classes are not evenly represented, accuracy alone is a misleading metric - a model that simply predicts the majority class will achieve high accuracy while being useless. Use precision, recall, and F1-score on a per-class basis, and report the **macro-averaged F1-score** as your primary comparison metric across models, as it treats all classes equally regardless of their size.

The appropriate response to class imbalance in image classification is **class weighting**, which penalises misclassification of minority classes more heavily during training. Keras supports this directly via the `class_weight` argument to `model.fit()`. The weights can be computed from the file counts in each class directory before the dataset is loaded.

Discuss in your paper whether class imbalance is present in your dataset, whether you addressed it, and what effect it had on per-class performance.

3.5 Additional Test

Take **4 photographs yourself** (can be of your own items) and test whether your final chosen model correctly classifies them. **Each student must submit their own photographs.**

Include a handwritten note with your name and student ID visible somewhere in each photograph. This confirms the images are your own.

For Garbage: photograph items around your home and test whether the model correctly identifies the waste category.

For Food: photograph food items and test whether the model correctly identifies them.

If the model gets some wrong, discuss why – is the image ambiguous? Is it a category the model consistently struggles with according to the confusion matrix?

3.6 Deliverables

1. **A paper** of at most 6 pages. Written in academic style (third person). No code in the paper – describe what you have done, not how the functions work. Submitted as PDF only. Use the provided template (.tex or .docx).
2. **A Jupyter notebook** with all code cells run and all outputs visible. This is submitted for verification of the results in your paper.
3. **Your 4 photographs**, submitted alongside the above.

3.7 AI Tools

You are responsible for every line of code and every claim in your report regardless of how they were produced.

If you used an AI tool, include a brief appendix (not counting toward the 6-page limit) stating: what you asked it, what it produced, and what you changed, rejected, or could not get to work – and why. This

appendix will not be marked, but its absence when AI use is evident will be treated as an attempt to misrepresent your work.

4 Expectations

- Conduct and document initial data analysis: class distribution, sample images, image dimensions, pixel value range.
- Do not fill the paper with descriptions of what CNN or transfer learning *are*. What matters is your work, your results, and your interpretation.
- Do not assume the reader is familiar with specific library functions. Describe the process, not the function call.
- Do not insert code into the paper. All code goes in the notebook.
- Excellent grammar, spelling, and technical writing are expected.
- Explain your findings thoroughly but concisely.
- Show evidence of learning and interpretation from the results, not just a list of numbers.
- If you need to describe an algorithm, use pseudocode.

5 Suggested Structure

You are not required to follow either of these exactly – use whatever headings best reflect your work.

1. Introduction
2. Data and Preprocessing
 - (a) Dataset Description
 - (b) Exploratory Analysis
 - (c) Preprocessing and Splits
3. Experiments
 - (a) CNN from Scratch
 - (b) Transfer Learning
4. Results and Comparison
5. Real-World Test
6. Conclusion
7. References

6 Requirements

- Paper must be submitted as PDF only. Other formats will lose 10%.
- 5% deducted per day late, in accordance with ATU Marks and Standards.
- 10% deducted per page over the 6-page limit.
- Plagiarised work receives 0% for a first offence.
- This is a solo project.

7 Plagiarism

Student plagiarism occurs when a student presents the work of another as their own without appropriate acknowledgement. It includes:

- Copying material from the internet, books, journals, or other sources without acknowledgement;
- Paraphrasing the work of others without acknowledgement;
- Submitting work written by somebody else, such as another student or a family member;
- Purchasing work from an internet site and submitting it as one's own;
- Presenting work produced collaboratively as individual work;
- Submitting the same piece of work for multiple assignments.

Plagiarism can be deliberate or accidental. Both are treated seriously.

8 Marking

The assignment is marked primarily on the paper. The notebook is submitted for verification of the results reported in the paper.

Marking categories:

- Overall Document Quality – writing style, presentation, structure
- Data Analysis and Visualisations
- CNN from Scratch
- Transfer Learning and Fine-Tuning
- Interpretation of Results
- Overall Results and their Presentation

Each category is marked on a 5-point rubric:

0	1	2	3	4	5
Not attempted / completely wrong	Poor	Weak	Average	Good	Excellent

Overall Results and Overall Document Quality are weighted more heavily than the individual technical sections.

9 Practical Notes

Google Colab. The easiest workflow is to store your dataset on Google Drive and access it from Colab. Upload the zip file to Drive, then unzip it from within the notebook. See <https://www.aboutdatablog.com/post/how-to-successfully-add-large-data-sets-to-google-drive-and-use-them-in-google-colab> for step-by-step instructions.

Memory issues. If you are running out of memory, the most likely cause is how you are loading the data. If you are not using `image_dataset_from_directory` with `prefetch`, start there. Also check `model.summary()` – if the parameter count runs to tens of millions, reconsider your image size or architecture. If memory is still tight, consider reducing the image size or applying stratified random sampling to reduce the dataset (stratified so that class proportions are preserved).

Workflow advice. Do not try to build and compare everything at once. Get one model working well first. Diagnose overfitting or underfitting. Fix it. Then build the next model. CNNs require you to work methodically.

References

- [1] Lukas Bossard, Matthieu Guillaumin, and Luc Van Gool. Food-101 – mining discriminative components with random forests. In *European Conference on Computer Vision*, 2014.
- [2] Suman Kunwar and Prashant Poudyal. The garbage dataset (GD): A multi-class image benchmark for automated waste segregation, 2026.